VLDB 2026 · INDUSTRIAL TRACK

Clock2Q+

Correlation-Aware Cache Replacement for Metadata Caches

Deployed in VMware vSAN

Yiyan Zhai[1] Bintang Dwi Marthen[2] Sarath Balivada[3] Vamsi Sudhakar Bojji[3] Eric Knauft[3] Jitender Rohilla[3]
Jiaqi Zuo[3] Quanxing Liu[3] Maxime Austruy[3] Junlong Gao[3] **Wenguang Wang**[3] **Juncheng Yang**[2]

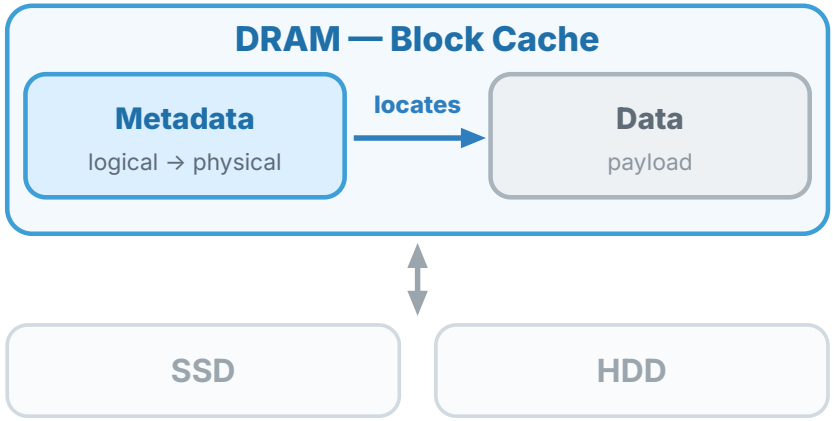
[1]Carnegie Mellon University [2]Harvard University [3]Broadcom

THE SETTING

What is a metadata cache?

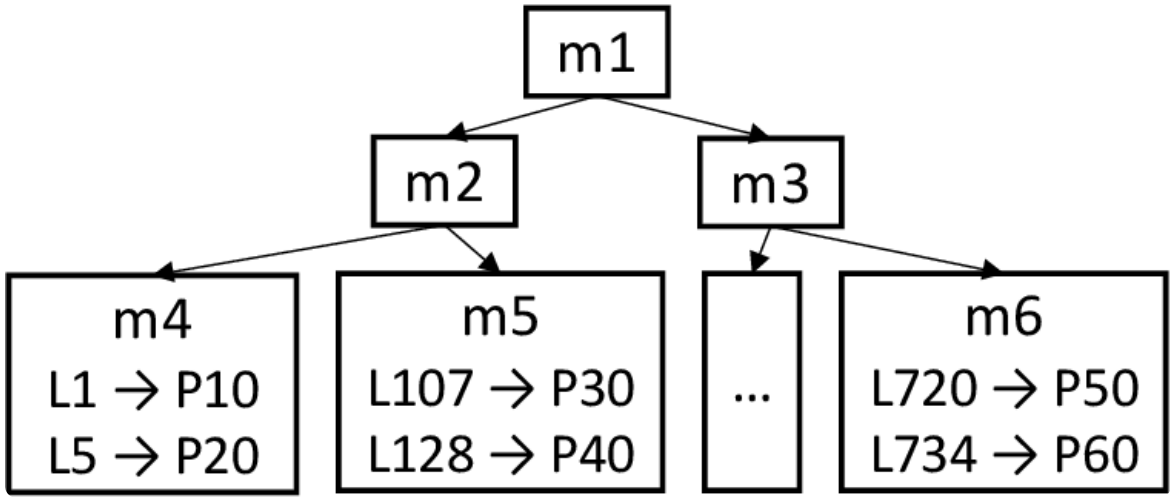
- **Data cache** = the payload.
- **Metadata cache** = the index that **locates** it. Maps logical addresses → physical addresses.
- The metadata cache is the **hot path** — touched on almost every access.

Judged not only on miss ratio, but also CPU overhead, scalability, and simplicity.



WHY METADATA IS DIFFERENT

Structural locality: a page packs many keys

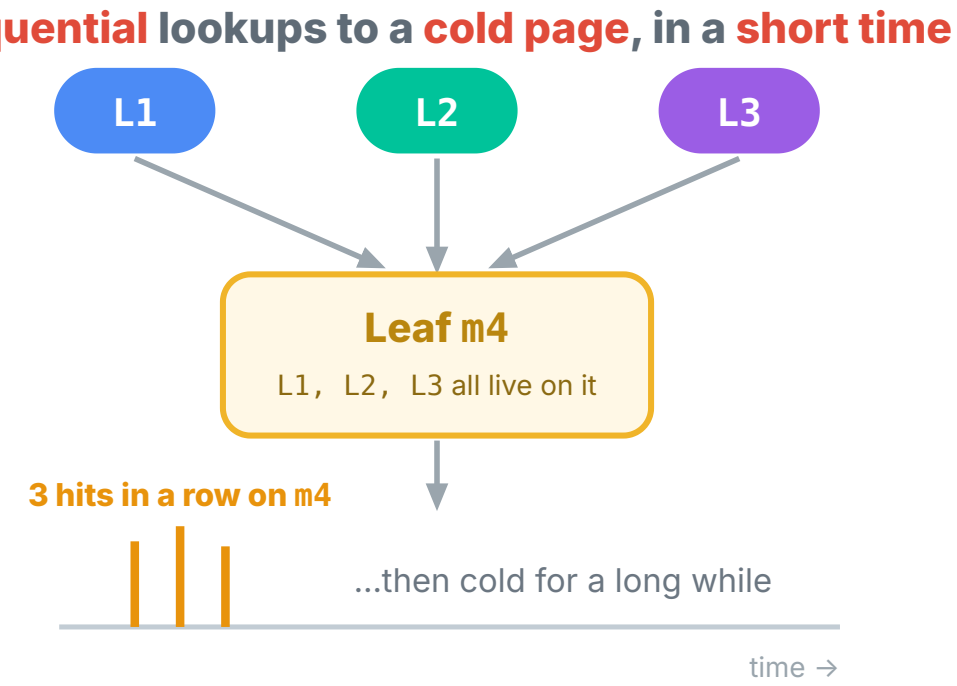


A **B-tree** maps **logical → physical** addresses; each node is one **metadata page**, packed with many mappings.

- A **metadata page** = a B-tree **leaf** holding **hundreds of key → location mappings** — not one key.
- A **sequential run of lookups** lands on the **same page** in a **short time**.
- The page *looks* hot — but that's **structure, not real popularity**.

THE PROBLEM

What is a correlated reference?

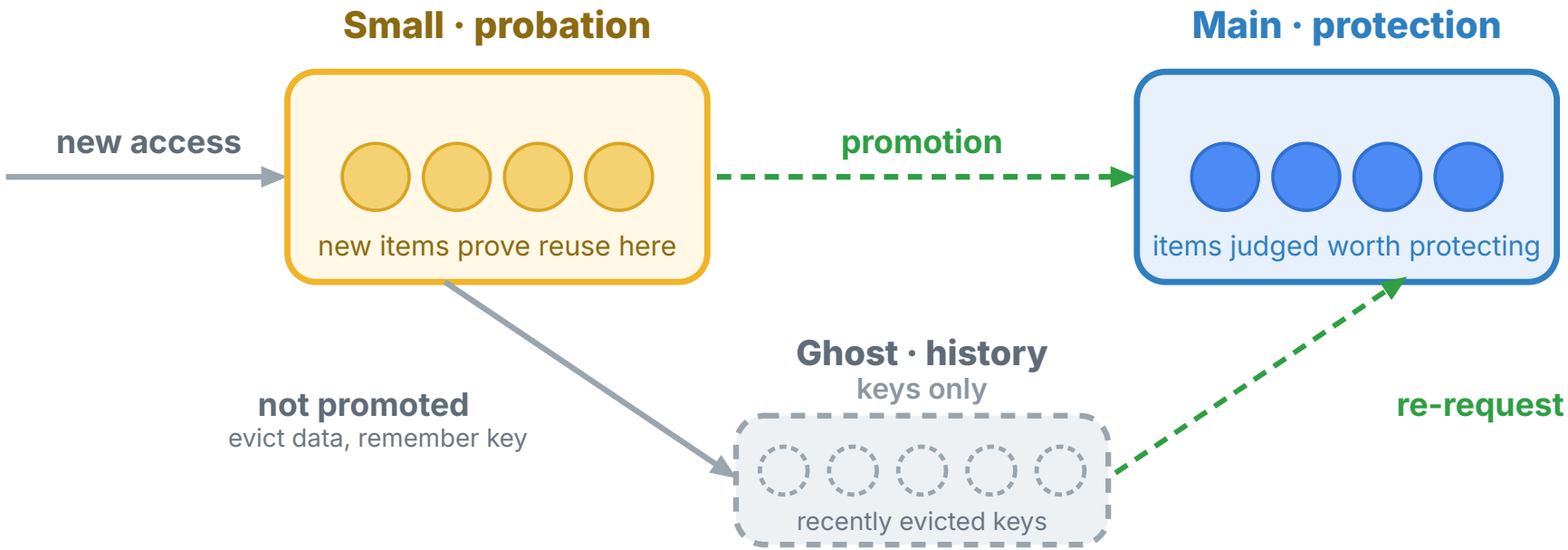


The trap: counting each hit as a vote for hotness. The page is busy by structure, not popularity.

BACKGROUND · THE SHARED DESIGN

These policies share a three-queue skeleton

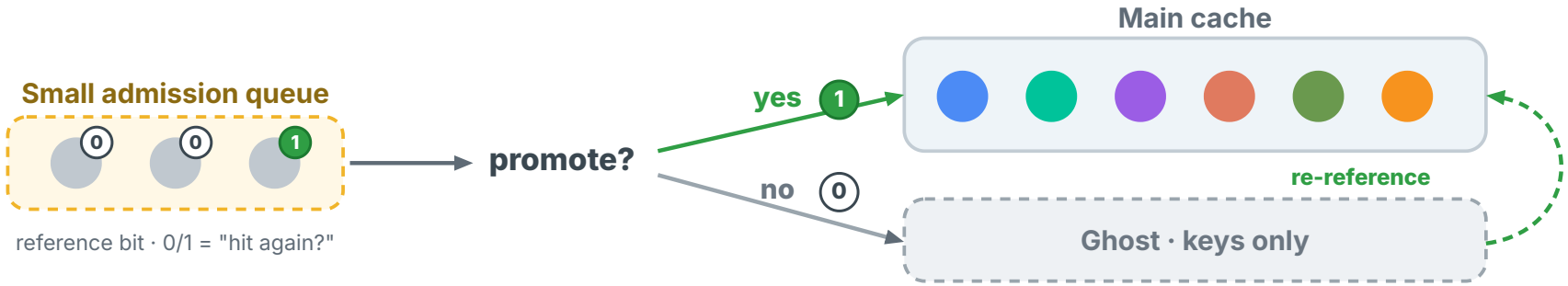
A new block must prove reuse before it occupies the protected cache.



The architecture stays; policies differ in how an item earns promotion to Main.

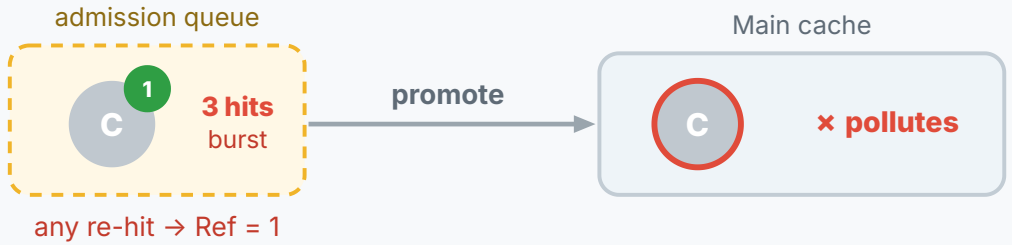
WHY EXISTING POLICIES FALL SHORT

Two opposite mistakes



S3-FIFO — admits cold blocks

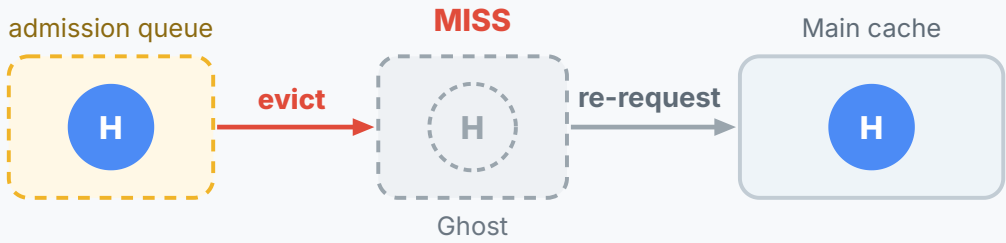
uses a **reference bit** · 0 / 1 = "hit again?"



a burst promotes a cold page

Clock2Q / 2Q — evicts hot blocks

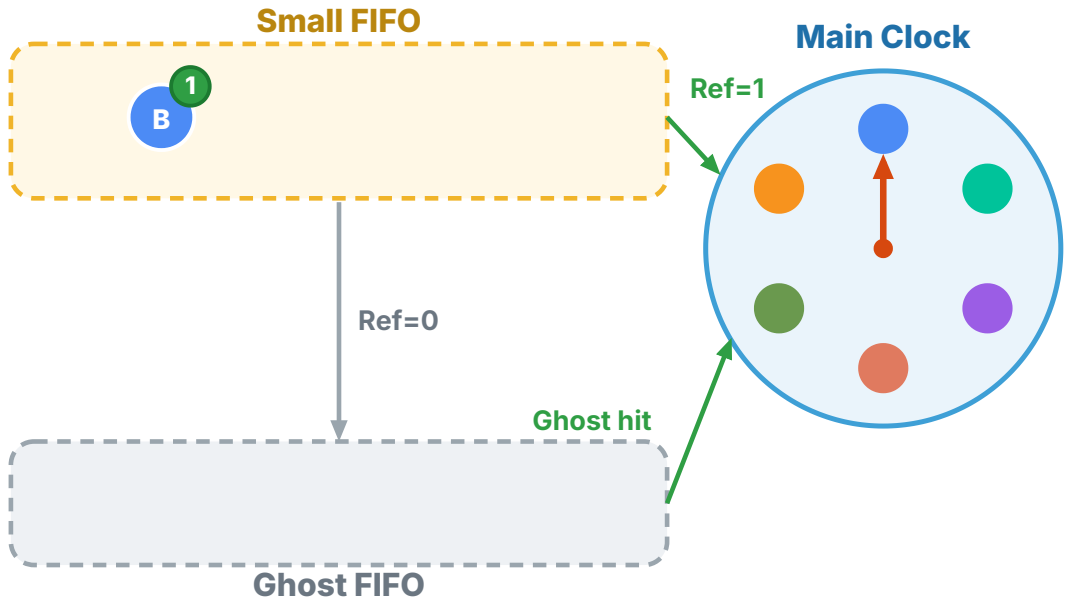
no reference bit — hot blocks must be dropped to Ghost



a hot block must miss once first

CLOCK2Q+

Insight: a correlation window as a filter



S3-FIFO

Where a re-hit happens tells you whether the popularity is real.

Hit inside the window | **Hit beyond the window**

S3-FIFO (window off): **any** re-hit → **Ref = 1** → even a burst pollutes the cache.

Clock2Q+ · inside the window: same burst → **Ref stays 0** → ages out to Ghost.

Clock2Q+ · beyond the window: real reuse → **Ref = 1** → promoted to Main.

METHOD — GENERATING METADATA TRACES

Turning public block traces into metadata traces

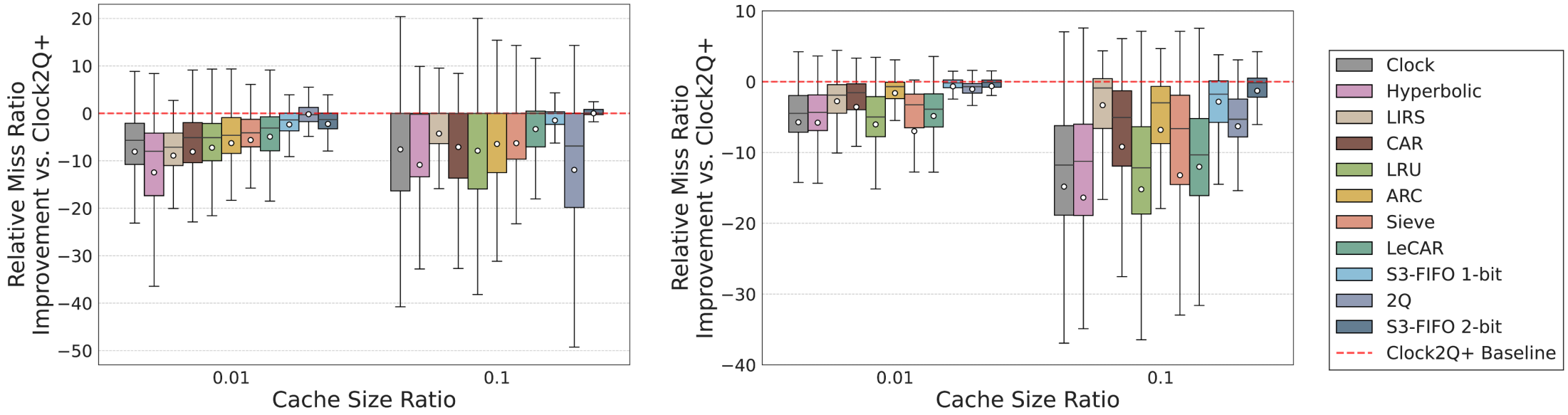
- No public metadata-cache traces.
- **Idea:** map each block to its leaf page: $\text{leaf} = \lfloor \text{block number} / \text{fan-out} \rfloor$ (fan-out = 200)
- Reproducible from **any** public block trace.

106	2.1B	libCacheSim	10
CloudPhysics traces	requests	simulator	SOTA baselines

RESULT — MISS RATIO

Lowest miss ratio on both metadata and data

Each box is a policy's miss ratio **relative to Clock2Q+. Below 0 = worse than Clock2Q+.**



Metadata traces. Higher is better.

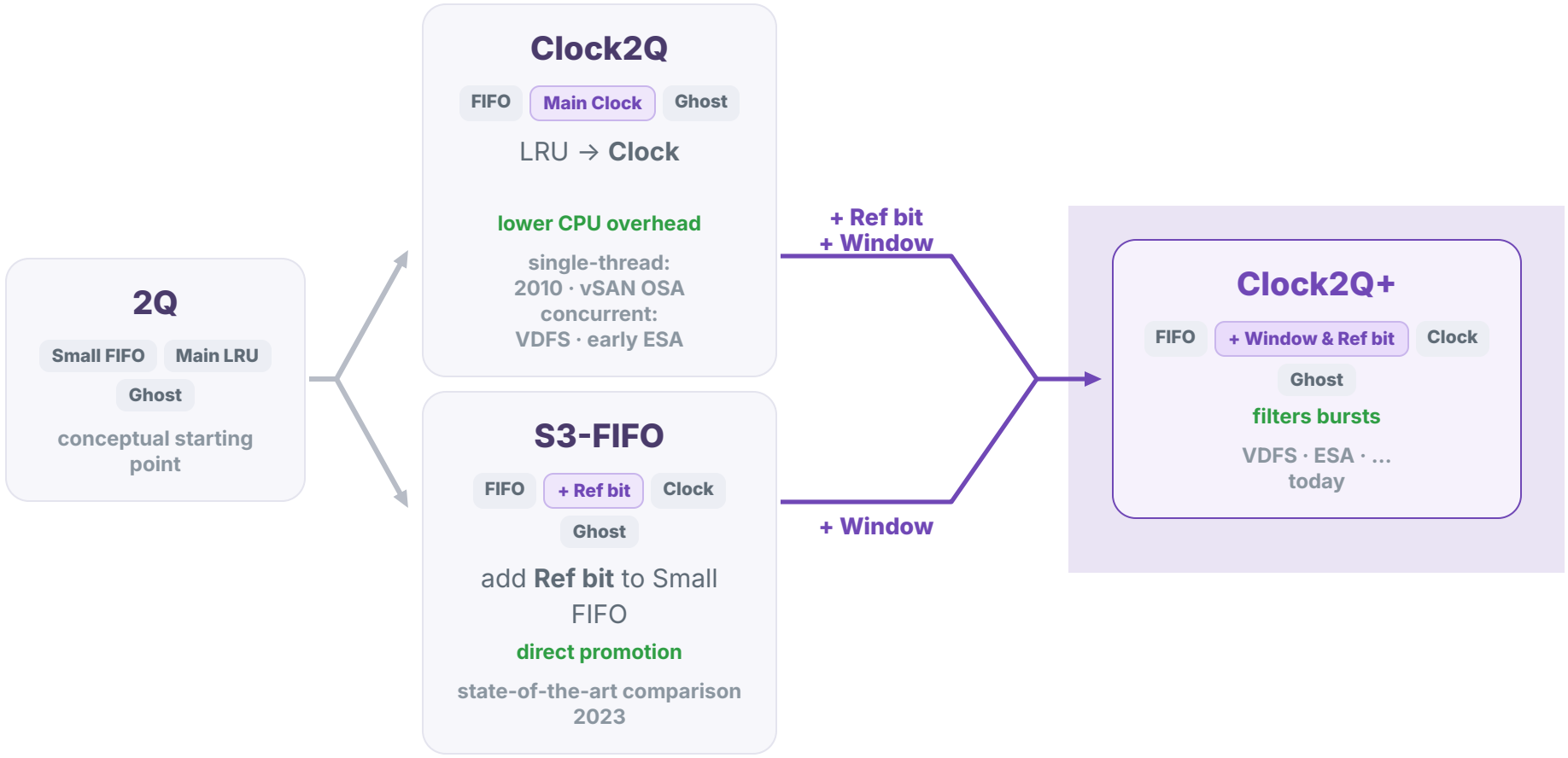
Data traces.

Best median & mean improvement at every cache size — up to 28.5% over SOTA on metadata.

In Juncheng Yang's course at Harvard, students build many policies — Clock2Q+ wins.

DESIGN LINEAGE

Over a decade of evolution



Simplicity — a first-class goal: every step stayed easy to implement, debug, and operate.

TAKEAWAYS

Clock2Q+

- **Correlated references** are a real, costly metadata pattern: a short burst makes a **cold** page look hot.
- The **correlation window** is a one-line change: filter bursts, still promote genuinely hot blocks directly.
- **Best miss ratio** among SOTA on metadata *and* data — and it is **deployed in VMware vSAN**.
- Low overhead, scalable, dirty-aware, resizable — **simplicity pays off** in production.

Reproducible in libCacheSim — Clock2Q+ & the metadata-trace method are upstreamed.

Clock2Q+

Correlation-Aware Cache Replacement for Metadata Caches · **deployed in VMware vSAN**

Get in touch

Yiyan Zhai (CMU) · **yiyanz@andrew.cmu.edu**

Wenguang Wang (Broadcom) · **wenguang.wang@broadcom.com**

Juncheng Yang (Harvard) · **juncheng@seas.harvard.edu**

Related sites

libcachesim.com — simulator

cachemon.org — caching-systems research

dataset.libcachesim.com — traces

python.libcachesim.com — Python API

github.com/1a1a11a/libCacheSim ↗

Carnegie Mellon University

BROADCOM

THANK YOU

ENGINEERED FOR PRODUCTION

Built to run safely at scale

⚡ Fast & lean

Array-based, allocation-free; the hit path is one hash lookup + a Ref-bit flip.

🧵 Scales on cores — the VIP requirement

LRU reorders on every hit → a write-lock per access, contention at scale. Clock's **Ref bit** is read-mostly: per-bucket mutex + atomic indices, no global lock.

🛡️ Robust under load

Skips dirty pages until flushed; capped scan per eviction — no CPU spikes, no infinite loops.

↔️ Resizes online

Grow / shrink the queues live via a background thread.

The bigger picture: textbooks teach LRU — but at multi-CPU scale its per-hit reordering doesn't hold up. Clock is what actually runs safely.

Simplicity — a first-class goal: simple data structures & no adaptive tuning.

BACKUP

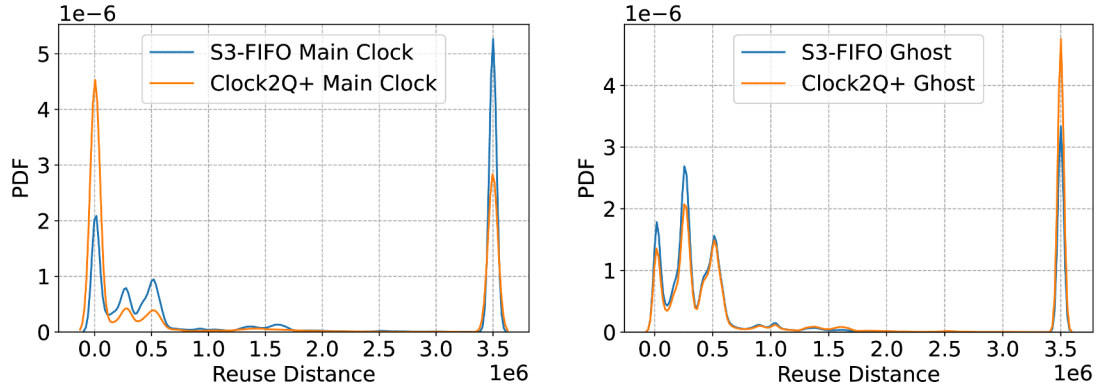
BACKUP — WHY IT WORKS

It promotes far fewer blocks — and the right ones

SMALL FIFO →	MAIN	GHOST
S3-FIFO	88,529	126,897
Clock2Q+	**20,762**	**186,959**

Clock2Q+ promotes < ¼ as many Small-FIFO blocks into Main.

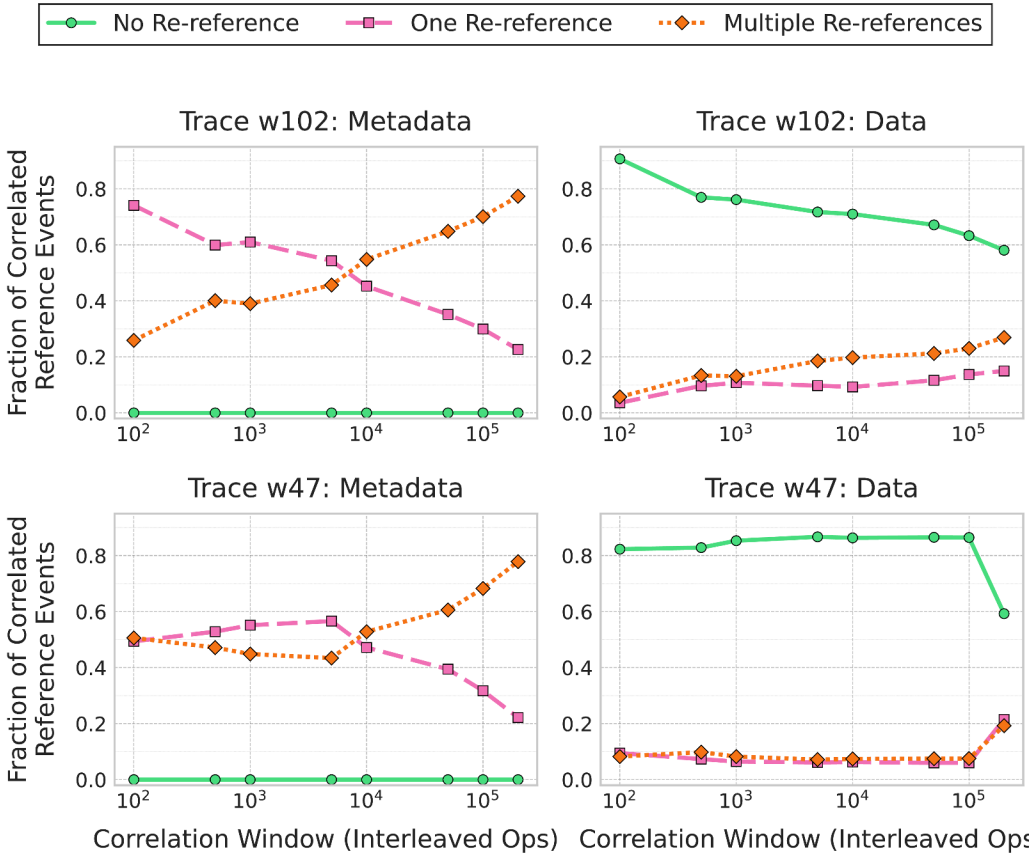
Correlated bursts are filtered out instead of polluting Main.



Promoted blocks have **short** next-reuse distance (hot); ghosted blocks have **long** distance (cold).

BACKUP — CORRELATED REFERENCES ARE REAL

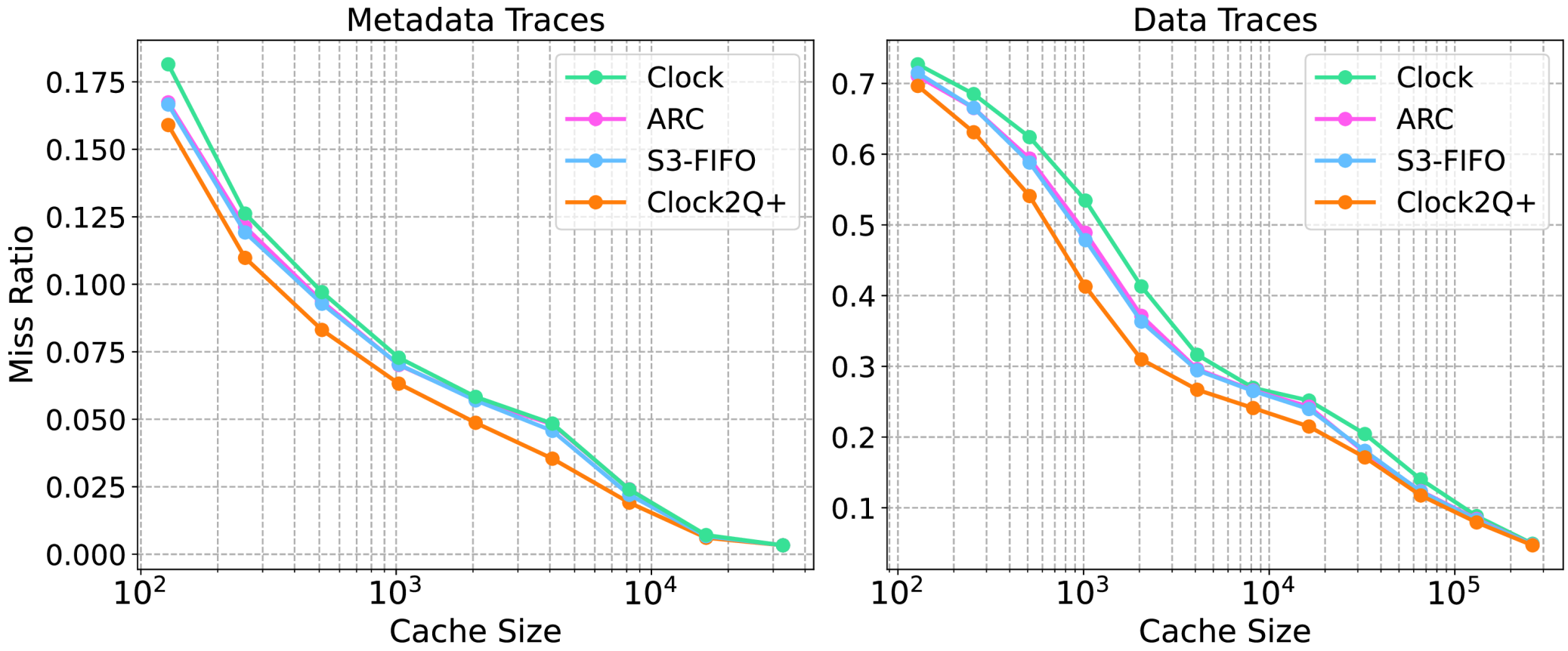
Bursts show up across production traces



Metadata pages get **multiple re-references** within a short window; data pages are **one-hit wonders**. (CloudPhysics w102, w47)

BACKUP — MISS RATIO VS. CACHE SIZE

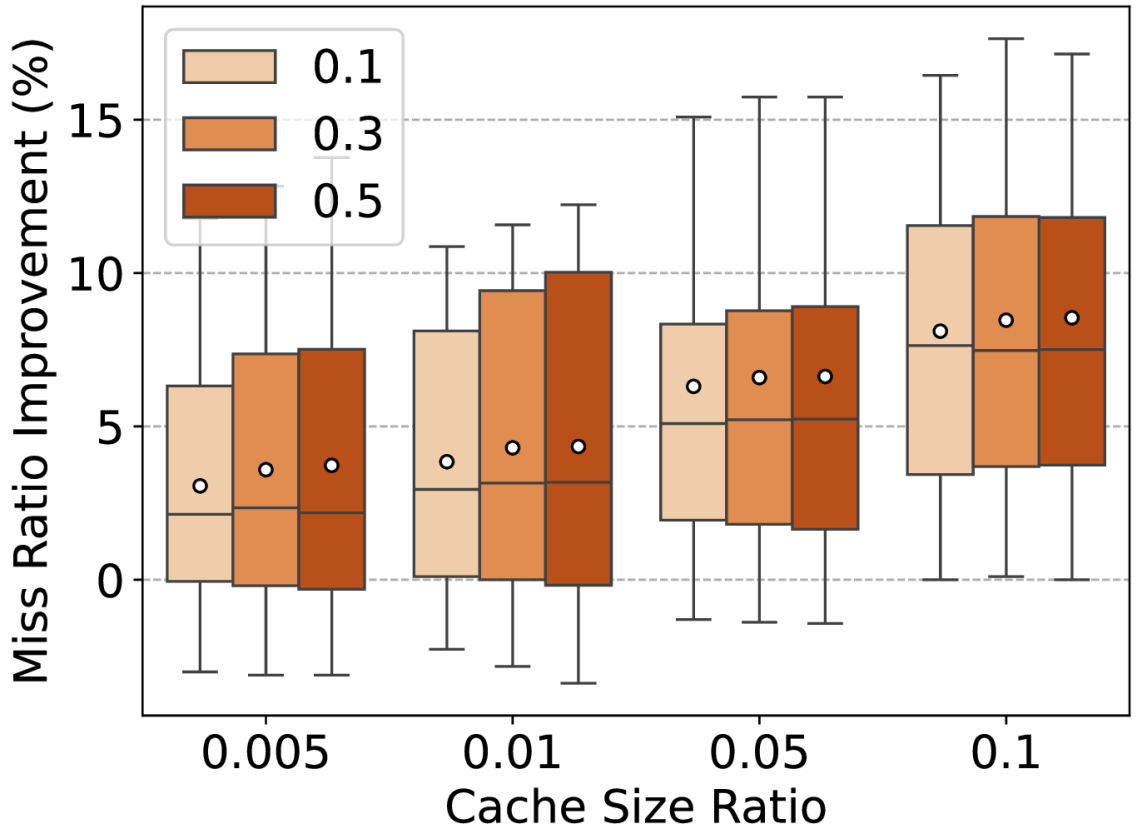
Lowest curve across the whole range



CloudPhysics w56. Clock2Q+ (orange) has the lowest miss ratio at every cache size, on metadata and data.

BACKUP — SENSITIVITY

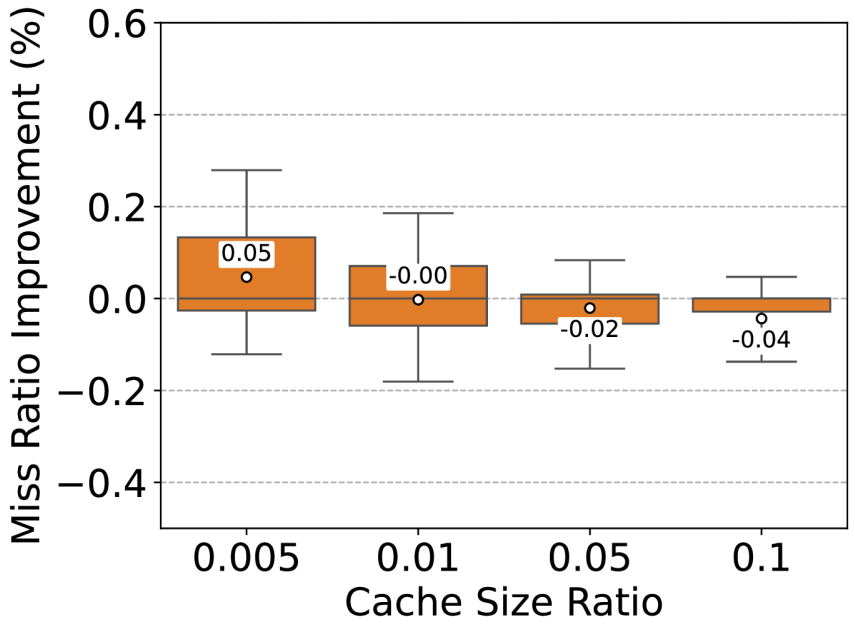
Robust to the window size (no tuning)



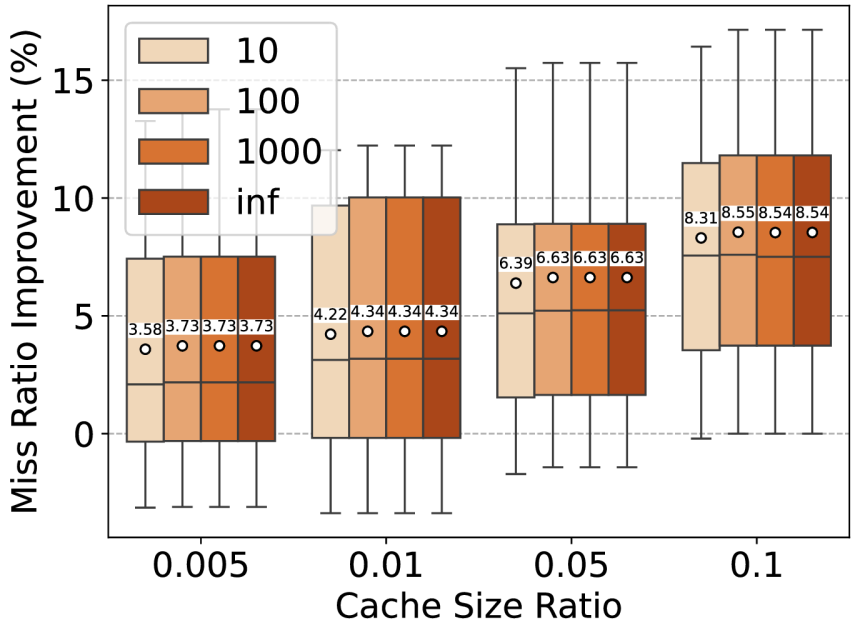
Correlation window = 10% / 30% / 50% of the Small FIFO — miss ratio is essentially unchanged.

BACKUP — PRODUCTION HARDENING

Dirty pages & bounded clock-hand work



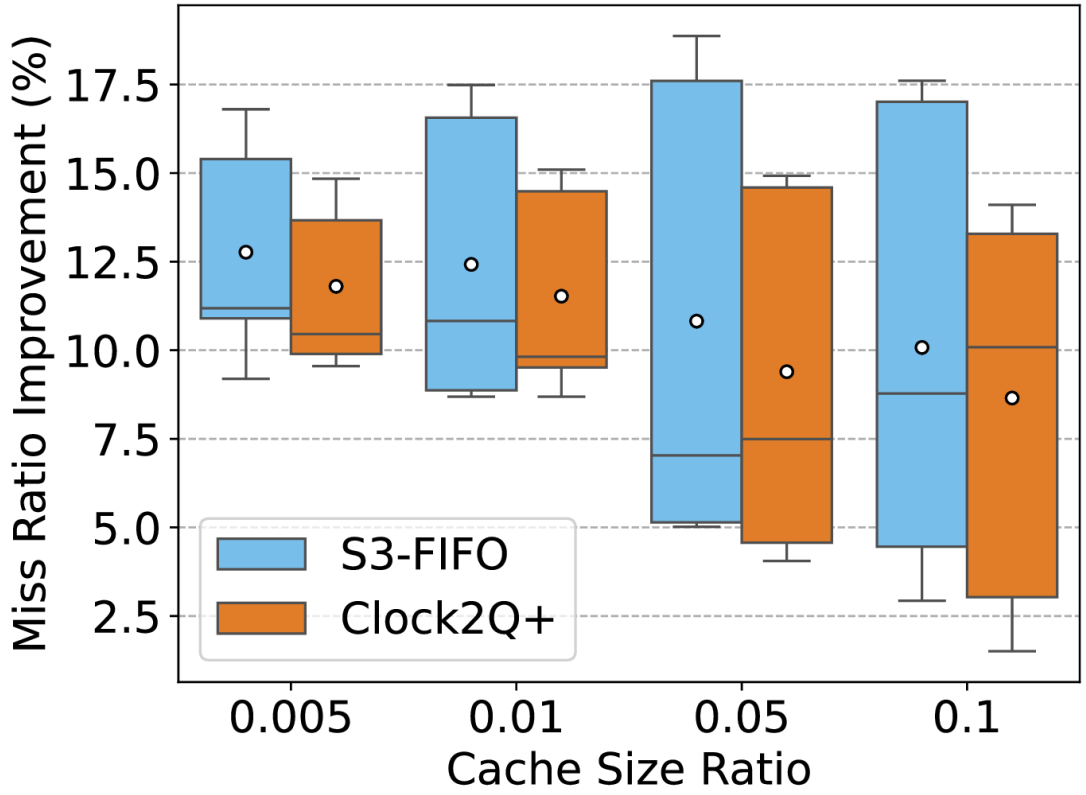
Simplified dirty-page handling: negligible miss-ratio impact.



Capping skipped blocks (limit = 10) bounds worst-case work safely.

BACKUP — LIMITATION

Where it doesn't win: non-block object caches



On Wikimedia / Meta KV / Tencent object traces (few correlated references), Clock2Q+ is on par with, and slightly behind, S3-FIFO.